Lab 09B: Multiple Factors and Post-Hoc Tests.

Question 1. Practice Building Models:

In this assignment, we will be working through the example in the textbook exploring an intervention in clinical psychology. In this hypothetical study, participants were assigned to receive either Psychotherapy (or not) and one of three Drug Treatment conditions (A, B, or Placebo).

In order to analyze these data, we will run a 3 x 2 factorial ANOVA in R.

- First, what is the Model A for the fully factorial model in this example?

$$\hat{y}_i = \beta_0 + \beta_1(P1_i) + \beta_2(D1_i) + \beta_3(D2_i) + \beta_4(P1 * D1_i) + \beta_5(P1 * D2_i)$$

The details of the ANOVA are presented below:

```
> Anova(mod1, type="III")
Anova Table (Type III tests)

Response: Mood
                   Sum Sq Df F value    Pr(>F)
(Intercept)          7200  1 1728.00 2.434e-14 ***
Drug                  453  2   54.36 9.647e-07 ***
Psychotherapy         450  1  108.00 2.359e-07 ***
Drug:Psychotherapy     57  2    6.84   0.01041 *
Residuals              50 12
```

- The F-value of 54.36 for the main-effect of Drug is a ratio of two different numbers, what are these two numbers?

54.36 = (453/2) / (50/12)

- Conceptually, what does this ratio represent?

The ratio of the mean squared error reduced compared to the mean squared error of the residuals is the error we have explained (per parameter) compared to the error we would expect to explain do to sampling variability alone.

- What is the most technically correct way to interpret the p-value for the main-effect of Drug?

If there was not a relationship between the type of drug administered and the level of symptoms (i.e., assuming the null hypothesis was true), we would very rarely ($p=9.6 \times 10^{-7}$) expect to explain this much variance due to chance.

- The proportional reduction in error for the Drug x Therapy interaction is 0.533. This PRE is a ratio of what numbers?

0.533 = (57) / (50+57)

- Note that the PRE values for individual effects do not add up to 1.00. Can you conceptually explain why these values to not sum to 1.0 even though the SSR + SSE(A) adds up to the SSE(C)?

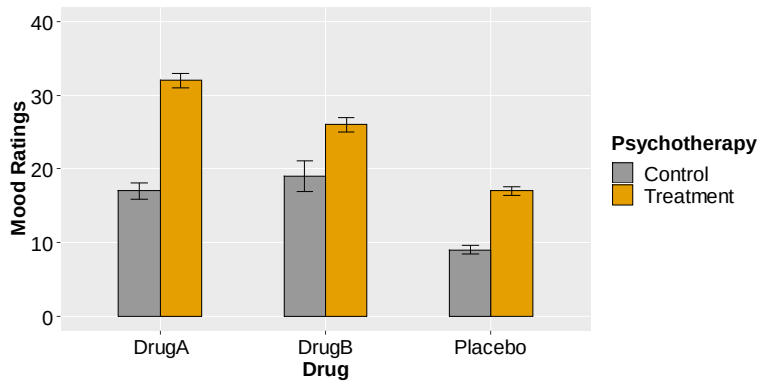
These values do not sum to one because the denominator in each fraction is different, representing the different Model A/Model C comparisons that go into the calculation of each effect.

E.g., the PRE for the Drug x Therapy interaction is 0.533 = (57) / **(50+57)** whereas the PRE for the main-effect of Drug is 0.901 = (453) / **(453+50)**.

Question 2. Post-Hoc Testing:

Because this factorial ANOVA has >2 levels per factor, we will need to conduct post-hoc tests to understand which groups are statistically different in our main-effects and interactions. Critically, we have a Drug x Psychotherapy interaction, which means that we can infer the effectiveness of the drug treatment depends on the psychotherapy that one is receiving. However, from the interaction alone, we do not know which of our groups are different from the others. The means, standard deviations, and sample sizes for each of the groups are shown in the table and figure below:

Drug	Psychotherapy	Mean Mood	SD	N
DrugA	Control	17	2.0000	3
DrugA	Treatment	32	1.7321	3
DrugB	Control	19	3.6056	3
DrugB	Treatment	26	1.7321	3
Placebo	Control	9	1.0000	3
Placebo	Treatment	17	1.0000	3



The omnibus test for the interaction tells us that the means of our groups are more different than we would expect from sampling variability alone. However, in order to discern which groups are different from others, we will need to conduct **post-hoc** tests. Post-hoc tests are so named because they come *after* the omnibus test.

There are different strategies for post-hoc testing that range from the more liberal to the more conservative in terms of controlling Type 1 Errors (false positives). For instance, in order to understand the interaction, we could re-run simpler ANOVAs with the effect of Psychotherapy in each Drug treatment condition. In these analyses, we can contrast the results of "uncorrected" post-hoc tests against a more conservative Bonferroni post-hoc test. In the uncorrected post-hoc test, we simply test the difference between each group (α = 0.05 for each test). For the Bonferroni post-hoc test, we adjust the alpha level by dividing by the total number of comparisons ($\alpha_{adjusted} = \alpha/c$, for each test).

- In our post-hoc test of the statistically significant interaction, how many comparisons do we need to conduct?
  9 (Compare therapies in each drug condition = 3, compare drugs in each therapy = 6)
- If your default α-level was 0.05, what would $\alpha_{adjusted}$ be if you wanted to run all of the possible post-hoc tests?
  0.05/9 = 0.0056

Question 3. Confidence Intervals:

Another way to show some of these differences between groups is with the 95% confidence intervals for each group's mean. The ***confidence interval*** is a method to estimate the parameters (true means in the population) from the means and variations we observed in the sample. Technically speaking, a 95% confidence interval means that if you repeatedly sampled from the same population, 95% of those samples' confidence intervals would contain the true mean. (This is a bit of a technical definition but note that this is not the same as saying there is a 95% chance that the true value falls in the confidence interval.)

Another equally valid interpretation of the confidence interval is that the limits (the upper bound and lower bound) are the most extreme values that are ***compatible*** with means we observed in our data. That is, it was unlikely (<5%) we would observe this mean if the true value was less than the lower limit or greater than the upper limit.

The confidence intervals like these can be drawn by multiplying the ***standard error*** (SE; the standard deviation for the sampling distribution of means) by the ***t-critical value*** (the t-value that separates a certain proportion of the sampling distribution from the rest.

Using the critical t-value and the SE, we can calculate the upper limits and lower limits for any given confidence interval via the formulas:

$$Upper\ Limit = \bar{x} + SE * t^*_{df,\alpha}$$

$$Lower\ Limit = \bar{x} - SE * t^*_{df,\alpha}$$

Here, we have 3 subjects in each condition, so if we wanted to estimate the mean in each condition our degrees of freedom would be **2** (N-1). The critical t-value that separates the middle **95%** of a distribution from the 5% in the tails is:

$t^*_{df,\alpha} = t^*_{2,0.025} = 4.303$

Note the t-value has an alpha of 0.025 because we want 5% of the distribution in both directions, so we only have 2.5% of the distribution in either tail (0.025 = 0.05/2). Use the table of means above and the t-critical value to fill in the table below:

			95% Confidence Interval	
Drug	**Therapy**	**Mean**	**Lower Bound**	**Upper Bound**
A	Control	17	12.0	22.0
A	Treatment	32	27.7	36.3
B	Control	19	10.0	28.0
B	Treatment	26	21.7	30.3
Placebo	Control	9	6.5	11.5
Placebo	Treatment	17	14.5	19.5

Next, you can also practice calculating a **99%** confidence interval using a more extreme t-critical value: $t^*_{df,0.005}$. Again note the t-value has an alpha of 0.005 because we want 1% of the distribution in both directions, so we only have 0.5% of the distribution in either tail (0.005 = 0.01/2).

$t^*_{df,\alpha} = t^*_{2,0.005} = 9.925$

Use the table of means above and the t-critical value to fill in the table below:

			99% Confidence Interval	
Drug	**Therapy**	**Mean**	**Lower Bound**	**Upper Bound**
A	Control	17	5.5	28.5
A	Treatment	32	22.1	41.9
B	Control	19	-1.7	39.7
B	Treatment	26	16.1	35.9
Placebo	Control	9	3.3	14.7
Placebo	Treatment	17	11.3	22.7

- In the simplest terms you can, explain why the 99% confidence intervals are wider than the 95% confidence intervals?
  Because the true mean is unknown, if we want to be more sure (i.e., confident) that we are capturing it, then we need to cast a wider net.